

Conceptual design

Requirements and Constraints:

In the conceptual design process, the main focus of S&C is to come up with an aircraft design that would satisfy both longitudinal and lateral stability. Among the requirements of S&C was to maintain the reasonable static margin that is between 5% to 15%. This also involves a minimal change in static margin when the payload is dropped. To achieve this, we had to come up with a design that would allow internal payload storage that is as close to the aircraft CG as possible. Other requirements and constraints included limiting the control surface deflection to 20 degrees to reduce drag and undesirable hinge moments. This can be achieved by proper sizing of the control surfaces. Another critical S&C issue during conceptual design was to come up with a design that would maintain stability in strong wind.

Goals

Maintain $C_{m\alpha}$ of less than 0

Minimize wing disturbances on the horizontal tail

Center of Gravity placement:

In order to ease the allocation of the center of gravity to assist in balancing the aircraft, the battery pack has been assumed to be movable along the fuselage length. The weight estimates of different components will be balanced laterally and longitudinally to place the CG in a required location.

Control surfaces:

Analysis and optimization was conducted in selecting the appropriate airfoil for the tail vertical and horizontal. Symmetric airfoils were chosen for horizontal and vertical tails as shown below.

Horizontal tail – NACA 0009

Vertical tail – NACA 0006

Preliminary design

In the Preliminary design process initial aircraft sizing and predicted aircraft weight was initiated, which is used to determine the lateral and longitudinal control performance of the aircraft. Trade studies are performed and optimization is done in order to come up with the best control surface sizing as possible. To ensure that the aircraft would be able to successfully complete the mission, both static and dynamic stability were analyzed. Control surface sizes, locations, and S&C derivatives are computed and optimized. Longitudinal and lateral-directional trim conditions for the mission.

Critical S&C parameters

- **Payload Configuration:** Payload must be located such that minimal CG shifts occur after the payload is dropped and does not adversely affect stability characteristics.

- **Static Margin:** To obtain sufficient maneuverability, the static margin (SM) must be kept at a low value while still maintaining adequate stability.

- **Twin Propulsion System:** Motors positioned on booms will create a thrust-induced yawing moment. Large thrust-pitch coupling requires aircraft CG to be near the thrust-line.

- **Control Surface Sizing and Placement:** The control surfaces have been sized to maintain lateral-directional control in high cross-wind. The technique of maximizing the distance from the CG to the longitudinal control surface AC to reduce control surface deflections was used. All control surfaces are sized to keep deflections under 20°.

Components weights and CG placement

All the weights of major components have been estimated using historical data. Table 1. Shows the summary of the weights and their respective distances from the leading edge. Aircraft components have been positioned in such a way that they put a resultant aircraft CG at a reasonable position. With the weight estimates, the aircraft CG is placed at 7.8 inches from the leading edge of the AC. This gives a static margin of 7.5%. The propulsion battery pack will be made to be movable to adjust the center of gravity so that the static margin can be flexible within a safety range of 5% to 20%. This will give enough tolerance in case the payload bay will be slightly shifted. The payload bay CG is made to be placed right by the CG of the aircraft.

Wing Sizing:

For a blended wing configuration, the wing occupies more space than any other component. The control surfaces of the wing have been sized using the Etkin and Raymer books. The aileron span has been assumed to extend to 50% of the wing span and 30% in chord.

.Aileron span is 13.8 inches

.Aileron chord is 1.8 inches

Tail and boom sizing sizing:

An H-Tail configuration with a high horizontal is the one that has been chosen for our design to keep it out of the wake at high angles of attack. The tail booms have been assumed to extend from the wing section. They are the same booms that are being used to support the props. The length of the boom is calculated using the method in Raymer. The moment arm (distance from the tail quarter chord to the wing quarter chord) which is part of the boom is calculated to be 39 inches which gave enough length to give a reasonable moment

Control surface sizing:

In order for the plane to have the required control in take-off and turn conditions, reasonable size of control surfaces is required. The control surfaces have been sized using Etkin and Raymer books .In order for a rudder to trim the aircraft for a sideslip at a particular velocity whether during cruise or stall, it has need sized to be 90% of the vertical span and 50% of the vertical chord. Also in order to trim the aircraft, the elevators have been sized to be 90% of the horizontal span and 50% of the horizontal chord

Horizontal tail

A high horizontal has been the preferable one so as to keep it away from the wing disturbances as possible .The horizontal tail span is the same as the distance between the two booms.

- Airfoil is NACA 0009
- Horizontal tail chord is 5.7 inches
- Horizontal tail span is 21.6
- Elevator chord is 2.75 inches
- Elevator span is 19.44 inches

Vertical tail

The distance between the verticals is the same distance between the two props because they are also sized by the booms.

- Airfoil is NACA 0006
- Vertical tail chord is 5.7 inches
- Vertical tail span is 7.1 inches
- Rudder chord is 2.25 inches
- Rudder span is 6.39 inches

S&C Preliminary predictions:

The methods outlined in Youtch and Morriss¹² together with detailed techniques in Roskam¹¹ were initially used to estimate the S&C derivatives and they are summarized in table 1. Later on the stability characteristic was evaluated using the vortex lattice method as implemented in AVL. The resulting derivatives obtained proved the plane to be stable in both static and dynamic conditions.

Table.2 shows the summary of the control derivatives

The trim plots are reflected in fig.1

Detail design

Aircraft Component Weight & CG Buildup

The weights associated with components used for the aircraft are maintained and updated regularly. Slight changes in the component weight made a slight difference in the aircraft CG but different components were moved around in order to maintain the reasonable SM. The new aircraft component CG buildup is presented in Table 3.

Wind Tunnel test and result analysis

Wind tunnel tests were run at different aerodynamic pressures and the data was analyzed and compared to the predicted values. There were slight offsets in the parameters due to different possible reasons. The change in lift coefficient slope increased the $C_{M-\alpha}$ to -0.004706. This is quite a huge swing but it still holds the stability of the airplane even at high angles of attack.

Pitching moment at zero angle of attack was found to be 0.00093 which is larger than the initially predicted value. One of the possible reasons here might be the inaccuracy in setting up the elevator deflection during model installation into the tunnel. Further different tests were run at the predicted elevator deflections during cruise and 1.2V stall and the pitching moment trend proved the stability of the airplane since it was negative all the way even though the pitching moment coefficients were not as exact as the predicted numbers.

The plots for the moment coefficients at different aerodynamic pressures for cruise and take off conditions are shown in fig.3

Cruise and take off Condition Analysis.

The model was tested at different dynamic pressures of 6, 16 and 24 to simulate cruise conditions. Elevator deflections were introduced to check trim conditions. The stall speed for the model was calculated to be 40.23 ft/s compared to 35.35 ft/s as predicted. The change in the value is because the model was tested at wide range of speeds with big intervals in between. Lesser speed intervals would have produced more accurate results. At higher speeds the model was able to generate enough lift with less elevator deflections. The pitching moment plot showed that the aircraft had a negative pitching moment (with a negative slope) the entire time indicating a stable flight.

For takeoff conditions, Different runs were done at dynamic pressures of 8 lb/ft.² and 5 lb/ft.²

Component	Weight (lbs)	x location (in)	Aircraft cg(in)	Item CG location relative to aircraft CG	moment about aircraft cg(lb.in)

Motor + Prop	0.80	-6.00	6.52	-12.52	-10.02
Wing	1.75	10.00		3.48	6.09
2 aileron servos	0.18	10.00		3.48	0.63
Nose gear(including wheels)	0.30	5.30		-1.22	-0.37
tail dreagger(including wheels)	0.03	43.84		37.32	0.93
Booms	0.25	21.55		15.03	3.76
Horizontal tail	0.08	43.84		37.32	2.99
Vertical tail	0.06	43.84		37.32	2.24
Rudder servo	0.06	43.84		37.32	2.09
Elevator servo	0.09	39.50		32.98	2.89
batteries	0.85	0.50		-6.02	-5.12
reciever/control rods	0.25	1.00		-5.52	-1.38
bomb bay servos	0.09	2.00		-4.52	-0.41
speed controller	0.16	1.00		-5.52	-0.88

Table1: components weights and their distances relative to the aircraft cg

$CL,0$	0.28	$CL,\delta e$	0.73	Cl,β	-0.17	Cl,p	-0.81	$Cn,\delta a$	-0.06
CL,α	3.82	$CM,\delta e$	-1.13	$Cn,\delta r$	-0.092	Cn,p	-0.59	Cy,r	0.97
$CM,0$	0.0962	Cy,β	-0.42	$Cl,\delta r$	-0.03	$Cy,\delta a$	0	Cl,r	-0.35
CM,α	-0.38	Cn,β	0.14	$Cy,\delta r$	0.19	$Cl,\delta a$	0.65	Cn,r	-0.09

Table 2 Static and dynamic stability derivatives

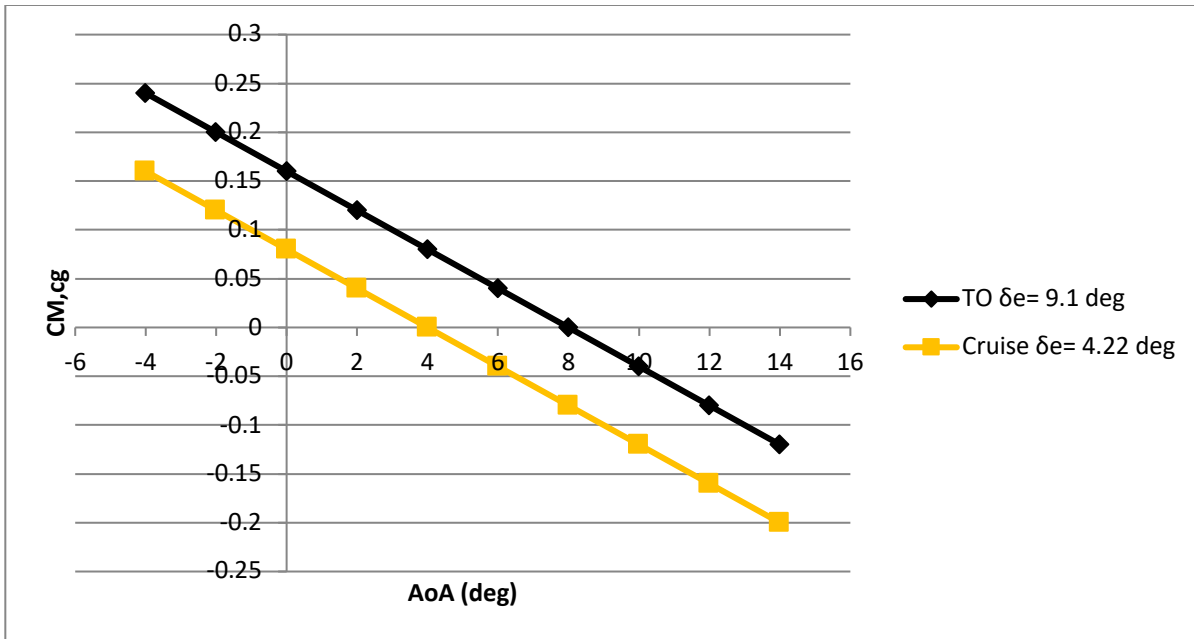


Fig.2 trim plot for cruise conditions

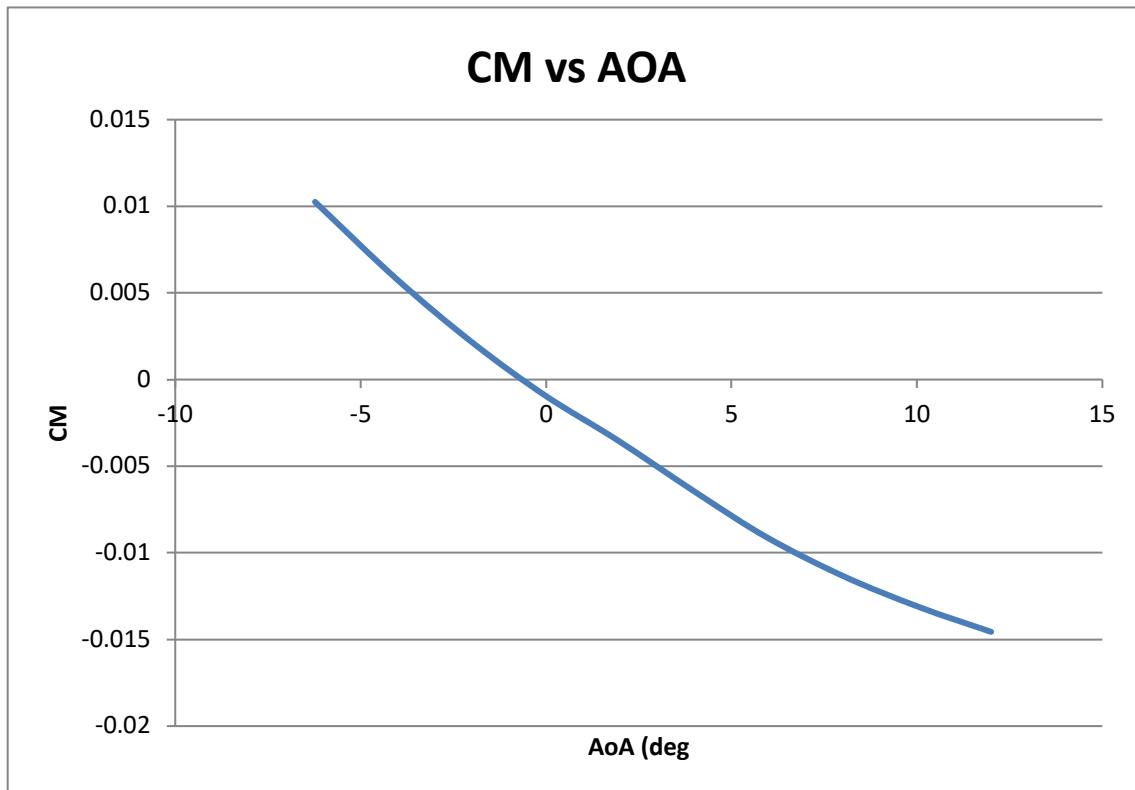


Fig.3 Wind tunnel data trim plot

Component	Weight (lbs)	x location (in)	Aircraft cg(in)	Item CG location re	moment about
Motor + Prop	0.40	-3	6.43696	-9.44	-3.774784
Wing	1.63	8.78		2.34304	3.8191552
2 aileron servos	0.18	8.78		2.34304	0.4217472
Nose gear(including wheels)	0.26	5.3		-1.13696	-0.2956096
tail dreagger(including wheels)	0.025	43.84		37.40304	0.935076
Booms	0.37	0		-6.43696	-2.3816752
Horizontal tail	0.08	43.84		37.40	2.9922432
Vertical tail	0.06	43.84		37.40304	2.2441824
Rudder servo	0.056	43.84		37.40304	2.09457024
Elevator servo	0.0875	39.5		33.06304	2.893016
batteries	0.95	0.5		-5.93696	-5.640112
reciever/control rods	0.25	1		-5.43696	-1.35924
bomb bay servos	0.18	2		-4.44	-0.7986528
speed controller	0.3	1		-5.43696	-1.631088

Table.3 refined weight build up and CG estimate